

Julie E. Walker

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Education

PhD Candidate in Interdisciplinary Ecology (GPA: 4.0, EGD: August 2021)

University of Florida, Gainesville, FL

Dissertation: Effects of Climate Change on Coastal Wetland Ecosystems and the Implications for their Future Management

Advisor: Christine Angelini (Christine.Angelini@essie.ufl.edu), Todd Osborne (Osbornet@ufl.edu)

Committee members: Andrew Altieri, Charles Martin

Bachelor of Arts in Biology (2013)

St. Mary's College of Maryland, St. Mary's City, MD

Graduated Cum laude with a minor in Environmental Studies, and Museum Studies

Research and Professional Experience

Bridging the Americas Marine Conservation Fellow (2016 – Current)

Department of Interdisciplinary Ecology, University of Florida, Gainesville, FL

- Joint fellowship between the University of Florida and the Smithsonian Tropical Research Institute to study the effects of global climate change on mangrove ecosystems across the Americas
- Extensively researched ecological implication of mangrove range expansion into temperate saltmarshes, as well as review current policy related to mangrove management
- Developed and implemented field and mesocosm methods for researching ecosystem functioning within the intertidal
- Published scientific research to peer reviewed journals, and presented work to multiple audiences
- Mentored undergraduate students and citizen scientist in fieldwork and video analysis
- Collaborated on multiple interdisciplinary research projects including University of Florida's School of Natural Resources, Environmental Engineering, Soil and Water Science, UF Law School, as well as the Smithsonian Tropical Research Institute

Environmental Management Staffer (2013 – 2016)

Chesapeake Bay Program, Annapolis, MD

- Facilitated and organized collaboration amongst the Fostering Chesapeake Stewardship Goal Team consisting of over 90 members representing NGO, State, Federal partners, and program leadership to achieve the Chesapeake Bay Program's Environmental Literacy, Public Access, and Land Conservation, and Citizen Stewardship goals.
- Assisted with managing Public Access and Land Conservation Datasets to track progress towards the goals outlined in the watershed agreement and created GIS maps for communication tools
- Assisted with the development of the Management Strategies and Workplans related to the Stewardship Goals and Outcomes.
- Managed and updated environmental education resource website (BayBackpack.com), added to existing resources, enhance user experience, and developed content. Promoted the site through presenting at educator conferences, writing for the blog, and maintaining media outlets
- Participated in federal grant panel review for watershed improvement and environmental education funds, and participated in interagency funding proposal for land conservation

Marine Ecology Intern (2013)

Marine & Estuarine Ecology Lab, Smithsonian Environmental Research Center, Edgewater MD

- Researched the effect of diel cycles of pH and DO on adult and juvenile oysters' growth and feeding as well as a possible link to Dermo disease contraction through mesocosm laboratory study.
- Maintained lab equipment, set up experiments, monitored tank chemistry.

- Managed oyster growth and disease contraction data and conducted statistical and graphic analysis. Presented results to Smithsonian Community and the Public.

Estuarine Biology / Education Intern (2012)

Calvert Marine Museum, Solomons Island, MD

- Developed and administered educational programs on the River Otters, and Touch Tank exhibit
- Assisted in the care of aquarium animals, including river otters, skates and rays, ctenophores, echinoderms, and fish, including preparing food and cleaning tank
- Monitored life support systems and water chemistry, implemented enrichment programs, and assisted in animal collection and quarantine.
- Interacted with visitors, giving informal discussions about exhibits

Lab Technician (2010-2011)

Chesapeake Biological Laboratory, Solomons Island, MD

- Researched the rate of sea level rising in the Chesapeake Bay area by examining environmentally sensitive Ostracod species' distribution in sediment samples
- Prepared the sediment samples, sorted, and identified Ostracod species
- Researched Scientific literature

Wildlife Biology Intern (2010-2011)

Patuxent River Naval Air Station, Patuxent River, MD

- Student Conservation Association position
- Conducted Forrest Interior Dwelling Species (FIDS) bird surveys to be used in land development to minimize the impact of construction on habitat sensitive bird species.
- Surveyed birds on the airfield to improve BASH (bird aircraft strike hazard) protocols.
- Assisted with other wildlife surveys such as emerald ash borers, wild indigo, deer populations, and reptiles and amphibians.
- Performed educational programs on biohazards for local schools and summer camps

Publications

1. **Walker, J. E.**, Angelini, C., Safak, I., Altieri, A. H., & Osborne, T. Z. (2019). Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. *Diversity*, 11(11), 208.
2. **Walker, J. E.**, Meyer, C., Barchiesi, S., Ankersen, T., Altieri, A.H., Osborne, T.Z., Angelini, C. Mangroves as a model system for managing common resources of international importance. *In prep*
3. **Walker, J. E.**, Osborne, T. Z., Martin C., Altieri, A. H., & Angelini, C., Influence of climate mitigated shifts in wetland vegetation on prey fish habitat use, behavior, and predation avoidance strategies. *In prep*

Selected Presentations

1. **Walker, J.E.**, Angelini, C., Safak, I., Altieri, A.H., Osborne, T.Z. Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. Science by the Shore, Marineland FL, September 2019. Oral Presentation.

2. **Walker, J.E.**, Angelini, C., Safak, I, Altieri, A.H., Osborne, T.Z. Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. Coastal and Estuarine Research Federation, Mobile, AL, November 2019. Oral Presentation.
3. **Walker, J.E.**, Angelini, C., Safak, I, Altieri, A.H., Osborne, T.Z. Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. North Florida Marine Science Symposium, Marineland, FL, January 2020. Oral Presentation.
4. **Walker, J.E.**, Angelini, C., Safak, I, Altieri, A.H., Osborne, T.Z. Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. The Water Institute, Gainesville, FL, February 2020. Oral Presentation.
5. **Walker, J.E.**, Angelini, C., Safak, I, Altieri, A.H., Osborne, T.Z. Effects of Changing Vegetation Composition on Community Structure, Ecosystem Functioning, and Predator–Prey Interactions at the Saltmarsh-Mangrove Ecotone. State of the Reserve, Ponte Vedra Beach, FL, February 2020. Oral Presentation.
6. **Walker, J.E.**, Angelini, C., Safak, I , Altieri, A.H., Osborne, T.Z. Mangroves on the Move. Mattie Kelly Environmental Symposium, Northwest Florida State College, Virtual, November 2020. Invited Speaker.
7. **Walker, J.E.**, Angelini, C., Safak, I , Altieri, A.H., Osborne, T.Z. Mangroves on the Move. Science on Tap, Florida Museum, Virtual, December 2020. Invited Speaker.

Awards

1. Florida Sea Grant Guy Harvey Award -\$5,000 (2019)
2. Emerging Leaders in Science Policy and Advocacy (ELISPA) – Science Policy Memo Writing Competition Winner (2019)
3. International Women's Fishing Association Award- \$500 (2019)
4. Coastal Estuarine Research Federation (CERF) student travel scholarship-\$200 (2013)
5. Maryland Saltwater Sport Fisherman Association (MSSA) Scholarship Award Winner- \$8,000 (2009-2012)

Teaching and Mentoring Experience

Guest Lecturer

- University of Florida Spring 2020, 2021; Course: Science Content for Elementary Educators, Topic: “Ecosystems, Food webs, and Niches”
- University of Florida Spring 2021; Course: Ecological Engineering, Topic: “Mangrove wetland management”

UF Multidisciplinary Research Experiences for Teachers, Instructor (2018)

- NSF funded Curriculum-based STEM Program for teachers, University of Florida
Personal Contribution: led presentations, lab and field activities, data analysis workshops, and development of elementary STEM classroom curricula for two local elementary teachers over two months.

Doris Duke Conservation, Graduate Student Mentor (2018)

- Mentored five undergraduate students, helped them develop independent research projects and trained them in field and laboratory techniques

Teaching Assistant - Ecology and Evolution (2013)

- St. Mary’s College of Maryland, St. Mary’s City MD
- Assisted students in Laboratory experiments
- Created and graded practical examinations and weekly quizzes

- Conducted review sessions
- Managed lab setup and inventory

Service

- Emerging Leaders in Science Policy and Advocacy (ELISPA)- Vice President (2020-2021)
- 500 Women Scientist Gainesville Chapter (2020- current)
- NSF Engineering Research Center Proposal, Student Internal Review Panel Member (2020)
- Emerging Leaders in Science Policy and Advocacy (ELISPA)- Advocacy Chair (2019)
- Ecological Society of America (ESA) Member (2020-current)
- Smithsonian Environmental Research Center, Marine Invasions Lab Volunteer (2015)
- NOAA Research Cruise, Clam Survey Volunteer (2015)
- International Science and Engineering Fair Special Awards Judge (2014, 2015)
- Deans Advisory Committee Member (2013)
- Atlantic Estuarine Research Society (AERS) Member (2010-2014)
- Daintree Rainforest Observatory, Queensland, Australia, Volunteer (2012)
- St. Mary's College of Maryland Student Ambassador (2009-2010)

Certifications and skills

- Laboratory Skills: Mesocosm design, video analysis, basic soil analysis, Basic GIS and R software knowledge
- Field Skills: Benthic crab identification, fish sampling methods, predation exclusion methods, community surveys
- Blog Management and Writing, Bay Backpack (2013-2016), Bridging the Americas (2016-Current)
- University of Florida Boating Safety Course (2021)
- Institutional Animal Care and Use Committee (IACUC)- completed application, standards and training (2018)
- NUI Scientific Diving Certification (Pending Dive Completion) (2021)
- ComSciCon 2020- Science Communication Workshop and Write-a-Thon (2020)
- Science on Tap Science Communication Professional Development Program (2020)
- NSPN Southern Science Policy Hub Conference and Training (2019)
- PADI Advanced Open Water Certification (2016)
- Leave No Trace Trainer Course (2015)
- Grant Writing and Researching Training for Non-Profits (2015)

References

- Christine Angelini, Assistant Professor in Environmental Engineering Sciences UF (Committee Co-Chair)
 - Phone: 352-294-7815
 - Email: c.angelini@ufl.edu
- Todd Osborne, Associate Professor in Soil and Water Science UF (Committee Co-Chair)
 - Phone: 904-461-4047
 - Email: osbornet@ufl.edu
- Andrew Altieri, Assistant Professor in Environmental Engineering Sciences UF (Committee Member)

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